



Senior Engineer, AI & Automation Platform — Vectera

Employment type:	Full-time
Level:	Senior Engineer — this role may ultimately be structured as a Founding Engineer position depending on the candidate; the recruiter should present it as open to both.
Location:	San Francisco, CA — or willing to relocate to San Francisco
Work authorization:	Candidates must have current work authorization in the United States
Company:	Vectera · Website: vectera.ai · LinkedIn: https://www.linkedin.com/company/vectera-ai/

About Vectera

Vectera is building AI-driven decision systems for institutional investors. Our platform combines machine intelligence, financial modeling, and large-scale investment datasets to support capital allocation across global markets.

Vectera is backed by a 40-year-old private equity firm with oversight of more than \$260 billion in capital, and draws on that firm's decades of proprietary investment data. The company is well-capitalized with no near-term need to raise external funding, and already has a roster of prospective institutional clients eager to adopt its technology. Over the past 18 months the team has built ten tools that form the core of the platform's engine, now at various stages of testing by internal users — this is a role joining a system with real traction behind it, not a blank slate.

About the role

We're hiring an engineer to build and own the automation that turns firm and client documents — spreadsheets, PDFs, decks, committee logs, due-diligence materials — into audit-ready institutional deliverables, including quarterly client reports and investment-committee memos. Every number has to foot, every factual claim has to trace to a source, and a person signs off before anything reaches a client or committee.

You'll draw the line between deterministic code and model judgment, keep the model's output honest once it's in the loop (multi-pass generation, critique, citation audit, full-source verification), and take systems from a local prototype to something running unattended on a server, reproducibly.

What you'll do

AI-assisted document automation

- Build ingestion pipelines across heterogeneous sources (multi-sheet workbooks, PDFs, decks with chart data trapped in raster images) into structured, verified output.

- Keep computation, ranking/selection, reconciliation, and compliance checks deterministic; use a model only where judgment is genuinely required, with its output grounded and verified, not trusted.
- Design conservative, traceable entity resolution — not silent fuzzy-matching.
- Handle missing, malformed, or irrelevant inputs gracefully: fail loud and safe, and distinguish "flag and continue" from "stop for human review."

Multi-pass generation & knowledge systems

- Build generate → critique → revise → audit → verify pipelines, not single-shot prompts.
- Verify absence-claims against the full source corpus, not just retrieved context.
- Keep domain/analytical knowledge in structured, human-editable data files, not hardcoded logic — the engine stays generic; a different knowledge pack should change the output, not the code.
- Design prompt systems as maintainable, composable scopes rather than one large fragile template.
- (*Stretch*) Contribute to systems that learn that structure from example documents rather than having it hand-authored.

Portfolio & data systems

- Contribute to the engines behind investment decisions — benchmarking, forecasting, and portfolio optimization/underwriting logic that turns constraints and forecasts into ranked recommendations.
- Work with structured, large-scale investment data (fund, deal, property, and portfolio-level) that's linked across levels rather than siloed, and help keep it that way as new tools consume it.

Human-in-the-loop & traceability

- Design the draft/approve boundary and how sign-off gets recorded.
- Make every material figure and factual claim resolve to a machine-readable source.

Platform engineering

- Own reproducibility and adaptability: same inputs → same output; next period's inputs should work via configuration, not a rewrite.
- Take systems from local scripts to server-deployed production — environment parity, secrets, scheduling, monitoring for unattended runs.
- Cache expensive third-party extraction calls; re-run only when sources actually change.
- Keep components isolated, with a single auditable handoff between them.
- Build a lightweight operator interface (kick off a run, review the output), and bring real testing, dependency management, and LLM-call observability.

Cross-functional

- Translate reporting and analytical rules from investment teams into code and verification logic.

What we're looking for

- **The ability to cleanly separate probabilistic AI behavior from deterministic business logic and validation** — knowing what must never be delegated to a model, and building the guardrails that keep it that way.
- Strong engineering fundamentals; production data pipelines shipped, not just prototypes.
- Real experience building LLM-integrated systems with grounding/verification, and a clear sense of where models fail and how to design around it.

- Experience with multi-pass/self-critique LLM workflows, and the judgment to know when that's worth it.
- Experience extracting data from unstructured/semi-structured documents, including chart data that only exists as pixels.
- Demonstrated testing/verification judgment — how you know output is correct, how you'd catch a regression.
- Comfort taking a system from a laptop to unattended production (secrets, scheduling, monitoring).
- Clear, honest technical communication about what a system does and doesn't do.

Nice to have

- Asset management, real estate, or other regulated financial services experience.
- Vision-capable/multimodal model experience for image/scanned-document extraction.
- Third-party document-parsing/OCR API experience and cost-aware caching.
- Lightweight internal tool/UI building for non-engineering stakeholders.
- Data governance/audit/compliance exposure.
- Information-extraction or taxonomy-induction experience (clustering/normalizing inconsistent content).
- We run primarily on Azure — useful (not required) familiarity with any of: Azure OpenAI, Azure AI Search, Azure AI Document Intelligence, Azure Blob Storage, Azure Functions or Container Apps, Service Bus, Azure SQL or PostgreSQL, Key Vault, Microsoft Entra ID, Application Insights. Not expected to have used every service.

How we hire for this

A practical take-home exercise plus a live technical round, focused on engineering judgment rather than trivia — the differentiator is whether you understand what you built well enough to defend and adapt it live.

Working style

Vectera operates with the pace, ownership, and responsiveness of a small, high-output engineering team. Best suited to someone who moves quickly, works through ambiguity, and owns things from design through production operation. Important releases and business-critical deliverables may periodically require flexibility outside standard working hours, and the team interacts multiple times throughout the day. This is not a 9-to-5 role, and it is not a lifestyle-friendly remote arrangement.